Physics - Undergraduate Level Assessment 21 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A satellite of mass m orbits a planet of mass M in a circular orbit of radius R. The time required for one revolution is
   (a) independent of M
   (b) proportional to $m^(1/2)$
   (c) linear in R
   (d) proportional to $R^(3/2)$

2. The energy required to remove both electrons from the helium atom in its ground state is 79.0 eV. How much energy is required to ionize helium (i.e., to remove one electron)?
   (a) 24.6 eV
   (b) 39.5 eV
   (c) 51.8 eV
   (d) 54.4 eV

3. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?
   (a) 2
   (b) 250
   (c) 5,000
   (d) 10,000

4. The eigenvalues of a Hermitian operator are always
   (a) real
   (b) imaginary
   (c) degenerate
   (d) linear

5. The negative muon, mu$^-, has properties most similar to which of the following$?

   6. Electron

   7. Meson

   8. Photon

   9. Boson

True / False (10 marks)

Answer True or False

6. True or False: Electromagnetic radiation emitted from a nucleus is most likely in the form of gamma rays.
7. True or False: In metals, the conduction electrons form a degenerate Fermi gas, leading to a mean kinetic energy significantly higher than kT.
8. True or False: A helium-neon laser is the most effective choice for spectroscopy over a range of visible wavelengths.
9. True or False: The angular magnification of the refracting telescope is 6 when using an eye-piece lens with a focal length of 20 cm.
10. True or False: An observer moving at 0.50 c parallel to a 1.00 m rod will measure its length as 0.80 m due to relativistic effects.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. An airplane traveling at a constant speed of 100 m/s drops a payload. Analyze the motion of the payload 4.0 s after release, focusing on its velocity relative to the airplane.
12. Discuss the implications of a reversible thermodynamic process, focusing on how it affects the entropy of the system and its surroundings.

End of Paper